

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 — SUB-STRAND 2.1: INTRODUCTION TO SALTS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have spent 8 lessons investigating salts — what they are, how they are classified, how they are prepared, and how they behave in the environment. Now it is time to bring everything together. Your task is to write a Final Explanation that fully answers the driving question:

"Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same — and what does understanding salts tell us about how to prepare and use them safely in Kenya?"

Your response must be written in continuous, well-organised prose — not bullet points. Use scientific vocabulary you have learned in this unit, support every claim with evidence from your investigations, and connect your explanation back to the real-world phenomenon you observed in the school storeroom. Aim for 4–5 detailed paragraphs. A strong response will show that you understand not just WHAT happened to each salt, but WHY — based on the structure, classification, and properties of salts.

Section 1: What Are Salts and How Are They Classified?

Begin by explaining what a salt is and how salts are classified. In your explanation, identify which class each of the three substances from the phenomenon belongs to — calcium ammonium nitrate (CAN fertiliser), sodium carbonate decahydrate (washing soda), and sodium chloride (table salt). Use the definitions and classification criteria you learned in this unit to justify your answers.

A salt is an ionic compound formed when the hydrogen ions (H^+) of an acid are wholly or partially replaced by metal ions or the ammonium ion (NH_4^+). Salts can be classified as normal salts, acid salts, or basic salts depending on how completely the hydrogen ions of the parent acid have been replaced. A normal salt contains no replaceable hydrogen ions and no hydroxide ions — for example, sodium chloride (NaCl) is a normal salt formed when all the hydrogen in hydrochloric acid is replaced by sodium. An acid salt still contains at least one replaceable hydrogen ion, such as sodium hydrogencarbonate (NaHCO_3). A basic salt retains hydroxide ions from the parent base.

All three substances in the phenomenon are salts. Sodium chloride (table salt) is a normal salt derived from hydrochloric acid and sodium hydroxide. Sodium carbonate decahydrate (washing soda, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) is also a normal salt of carbonic acid, but it crystallises with ten molecules of water of crystallisation. Calcium ammonium nitrate (CAN) is a mixture of calcium nitrate $\text{Ca}(\text{NO}_3)_2$ and ammonium nitrate NH_4NO_3 — both normal salts of nitric acid — widely used as a nitrogen-rich fertiliser on maize farms across the Rift Valley and other agricultural regions of Kenya.

Section 2: Water of Crystallisation, Hygroscopy, and Deliquescence — Explaining the Phenomenon

This is the heart of your explanation. Use the concepts of water of crystallisation, efflorescence, hygroscopy, and deliquescence to explain precisely what happened to each of the three salts overnight. Make sure you explain the difference between what happened to the washing soda and what happened to the CAN fertiliser, and why table salt was unaffected. Include any relevant chemical equations or formulas where they strengthen your explanation.

The three salts behaved differently because they have different interactions with water vapour in the air, and this is directly linked to their structure and properties.

The washing soda crystals ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) shrank and developed a white powdery crust because of a process called efflorescence. Washing soda contains ten molecules of water of crystallisation — water molecules that are chemically bonded within the crystal lattice and give the crystals their characteristic shape and size. When exposed to the relatively dry air in the laboratory, the vapour pressure of the crystals is higher than the partial pressure of water vapour in the surrounding air. As a result, the crystals spontaneously lose their water of crystallisation to the atmosphere. The white powdery crust that formed is anhydrous sodium carbonate (Na_2CO_3), which has a much smaller crystal structure than the hydrated form. The equation for this change is: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + 10\text{H}_2\text{O}(\text{g})$. This is why the crystals appeared to shrink — they lost mass and volume as the water of crystallisation escaped.

The CAN fertiliser granules clumped together and felt wet and sticky because calcium nitrate and ammonium nitrate are both deliquescent salts. A deliquescent salt absorbs water vapour from the air so strongly that it eventually dissolves in the absorbed water and forms a solution. This happens because the vapour pressure above these salts is lower than the partial pressure of water vapour in the humid storeroom air, so water molecules are drawn from the air onto the surface of the granules. The liquid you observed coating the CAN granules was not added water — it was water drawn directly from the air overnight. This is why the granules clumped: the absorbed moisture caused individual granules to stick to each other.

Table salt (NaCl), by contrast, remained completely unchanged. Sodium chloride is neither efflorescent nor deliquescent under normal Kenyan atmospheric conditions. Its crystal lattice is highly stable, it contains no water of crystallisation to lose, and its vapour pressure at room temperature is neither high enough to release water nor low enough to absorb significant moisture from the air. This is why NaCl is widely used in food preservation in Kenya — it is stable, does not cake easily under normal humidity, and does not change in composition when stored.

Section 3: How Salts Are Prepared — Evidence from the Laboratory

Explain the main methods used to prepare salts in the laboratory. For each method, state which types of salts it is used for and give one specific example using a salt relevant to the phenomenon or to Kenyan everyday life. Include relevant word equations or chemical equations. Explain why the correct preparation method matters — what could go wrong if the wrong method is chosen?

During this unit, you investigated several laboratory methods for preparing salts, and understanding these methods helps explain why salts have the properties they do.

The first method is neutralisation, in which an acid is reacted with an alkali (a soluble base) to form a soluble salt and water. For example, sodium chloride — the table salt in the phenomenon — can be prepared by reacting hydrochloric acid with sodium hydroxide solution: $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. The salt is then obtained by evaporating the water. This method is used for soluble salts where both the acid and base are soluble.

The second method is the reaction of an acid with a metal oxide or metal carbonate, which is useful for preparing soluble salts where no suitable alkali is available. For example, copper(II) sulphate can be prepared by adding excess copper(II) oxide to warm dilute sulphuric acid: $\text{CuO}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$. The excess copper oxide is filtered off, and the solution is evaporated and allowed to crystallise.

	The third method is direct combination or synthesis, used for preparing some salts directly from their elements. The fourth method is double decomposition (precipitation), used to prepare insoluble salts by mixing two solutions whose ions combine to form an insoluble product. For example, barium sulphate — an insoluble salt — can be prepared by mixing barium chloride solution with sodium sulphate solution: $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$. Choosing the wrong method matters because it can result in an impure product, an incomplete reaction, or even a hazardous situation. For instance, attempting to prepare an insoluble salt by neutralisation and evaporation would fail because the desired salt would precipitate out and become contaminated before pure crystals could form. In Kenya's agricultural context, understanding preparation methods also informs the quality and purity of fertilisers like CAN — impurities in fertiliser salts can affect crop uptake and soil chemistry.
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Section 4: Safety, Storage, and Application in Kenya

Using what you now know about salt properties — including hygroscopy, deliquescence, efflorescence, and solubility — explain the practical implications for how salts should be stored and used safely in Kenya. Give at least two specific, real-world examples (e.g., fertiliser storage, food preservation, laboratory chemical storage). What risks arise if salts are not stored correctly, and what precautions should be taken?	Understanding how salts interact with moisture in the air has direct and important practical consequences for Kenyan farmers, households, and laboratory workers. For farmers in the Rift Valley and across Kenya who use CAN fertiliser, the deliquescent nature of calcium nitrate and ammonium nitrate means that open or poorly sealed bags will absorb moisture from the humid air, causing the granules to dissolve, clump, and become difficult or impossible to spread evenly using a hand-spreader or machine applicator. Uneven application leads to over-fertilisation in some areas (which can burn crops and leach excess nitrates into waterways like the Mara River or Lake Naivasha) and under-fertilisation in others (reducing maize yields). To prevent this, CAN fertiliser bags must be stored in cool, dry conditions, sealed tightly after each use, and used within a reasonable time after opening. Farmers should never leave fertiliser sacks open in a humid storeroom overnight — as the phenomenon demonstrated. In the household kitchen, table salt (NaCl) is used to preserve foods such as nyama choma, dried fish from Lake Victoria, and vegetables. Because NaCl is neither deliquescent nor efflorescent under typical conditions, it remains stable and effective as a preservative. However, in very high humidity — such as in a coastal town like Mombasa during the long rains — table salt can absorb enough moisture to cake. Adding a small amount of an anti-caking agent (such as magnesium carbonate) prevents this. In the school laboratory, efflorescent salts like washing soda must be stored in airtight containers to prevent them from losing their water of crystallisation and changing mass. This is especially important when washing soda is used in titrations or gravimetric analysis — if the sample has partially effloresced, its molar mass will have changed, leading to inaccurate results. All hygroscopic and deliquescent chemicals must be labelled with appropriate hazard information and stored away from moisture sources. When handling ammonium nitrate (a component of CAN), care must also be taken because it is an oxidising agent and can be a fire hazard if contaminated with organic material.
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Section 5: Full Answer to the Driving Question

Write a final concluding paragraph (or two) that directly and completely answers the	You can now fully explain what you observed in the school storeroom that morning. The three salts — CAN fertiliser, washing soda, and table salt — behaved differently because their crystal structures, compositions, and interactions with atmospheric water vapour are fundamentally different, even though all three are ionic compounds formed from acid-base reactions.
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driving question. Your answer must connect the science of salts — classification, water of crystallisation, hygroscopy, deliquescence, efflorescence, and preparation — back to the original observation in the school storeroom. Show how your understanding has grown across the 8 lessons by explaining not just what happened, but why, and what this means for real life in Kenya.	The CAN fertiliser granules became wet and clumped because calcium nitrate and ammonium nitrate are deliquescent: they absorb water vapour from the humid storeroom air so strongly that they begin to dissolve in it, coating the granules in a sticky solution. The washing soda crystals shrank and formed a white crust because $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is efflorescent: the ten water molecules held within its crystal lattice escaped into the drier laboratory air, leaving behind the anhydrous white powder Na_2CO_3. Table salt remained unchanged because NaCl has a stable crystal lattice, contains no water of crystallisation, and does not absorb or release moisture under normal atmospheric conditions in Kenya. Understanding these differences is not just an academic exercise. It directly informs how Kenyan farmers should store fertilisers, how laboratory technicians should handle and weigh hydrated salts, how food-processing workers should manage salt-based preservatives, and how chemists choose the correct preparation method for a given salt. Across all eight lessons, you have built a model of salts that goes far beyond a simple definition — you can now classify salts, write equations for their preparation, explain their physical behaviour in terms of molecular structure and thermodynamics, and apply this knowledge to solve real problems in Kenyan agriculture, food science, and the laboratory. That is what it means to truly understand salts.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific Accuracy of Salt Classification and Definitions	Correctly defines salts as ionic compounds formed by replacement of H^+ ions; accurately classifies all three phenomenon salts (NaCl , $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$, CAN) with full justification referencing parent acids and bases; uses all key vocabulary (normal salt, acid salt, basic salt, water of crystallisation) correctly throughout.	Correctly defines salts and classifies at least two of the three phenomenon salts with adequate justification; uses most key vocabulary correctly with minor errors or omissions.	Attempts a definition of salts but it is incomplete or partially inaccurate; classifies only one phenomenon salt correctly or provides classifications without justification; key vocabulary is used inconsistently or incorrectly.
Explanation of Atmospheric Behaviour (Deliquescence, Efflorescence, and Stability)	Provides a precise and complete explanation of deliquescence (CAN), efflorescence (washing soda), and stability (table salt) using vapour pressure reasoning; includes correct chemical equation for efflorescence of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$; clearly explains WHY each salt behaves differently in terms of crystal structure and water of crystallisation.	Correctly explains the behaviour of all three salts and links each to the correct process (deliquescence/efflorescence/stability); may include the chemical equation; explanation of WHY is present but may lack depth on vapour pressure or crystal structure.	Identifies that the salts behave differently and uses at least one correct term (e.g., deliquescent or efflorescent) but explanation of the underlying reasons is missing, incomplete, or contains a scientific misconception.
Knowledge of Salt Preparation Methods	Describes at least three preparation methods (e.g., neutralisation, acid + metal oxide/carbonate, precipitation/double decomposition) with correct conditions; gives relevant examples including at least one linked to	Describes at least two preparation methods correctly with examples; includes at least one equation; explanation of method choice is present but may not connect clearly to the	Describes one preparation method with limited accuracy; equations are missing or incorrect; no clear connection to the phenomenon or real-world context.

	the phenomenon or Kenyan context; includes correct word or symbol equations; explains why method choice matters.	phenomenon or Kenyan context.	
Application to Kenyan Real-World Contexts (Safety and Storage)	Provides at least two detailed, accurate, and specific real-world applications rooted in Kenyan contexts (e.g., CAN fertiliser storage in the Rift Valley, table salt preservation of fish from Lake Victoria, laboratory handling of efflorescent salts); clearly links each application to a specific salt property studied in the unit; identifies concrete risks and precautions.	Provides two real-world applications with links to Kenyan contexts; connections to salt properties are mostly accurate; risks or precautions are mentioned but may lack specificity.	Provides one real-world application or applications that are generic and not clearly linked to Kenyan contexts or specific salt properties studied in the unit.
Coherence, Structure, and Use of Evidence to Answer the Driving Question	The final explanation directly, completely, and convincingly answers all parts of the driving question; all claims are supported by evidence from investigations; the response is written in well-organised prose with clear logical flow across sections; scientific vocabulary is used accurately and consistently; the response explicitly connects back to the original storeroom phenomenon in the conclusion.	The explanation answers most parts of the driving question with adequate evidence; prose is organised and mostly clear; scientific vocabulary is mostly accurate; some connection back to the original phenomenon is made in the conclusion.	The explanation partially addresses the driving question but key parts are unanswered; evidence is limited or not clearly linked to claims; response may rely on bullet points or lists rather than well-developed prose; connection to the original phenomenon is weak or absent.